

Development of an Integrated Smart Work Zone for Proactive Intrusion Detection and Warning – Digital Twin

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1. INTRODUCTION

- **Work zone intrusions** have been recognized as a safety-critical issue in the U.S.
- Work zone intrusion is a **multi-dimensional** issue, driven by a combination of **worker-related**, **vehicle-related**, and environmental factors

Across the U.S. (2022)		Within the Indiana (2022)	
	820 work zone fatal crashes		6348 collisions within work zone
	890 fatalities caused by crashes		30.7 fatalities rate / 1000 collisions

2. RESEARCH OBJECTIVES

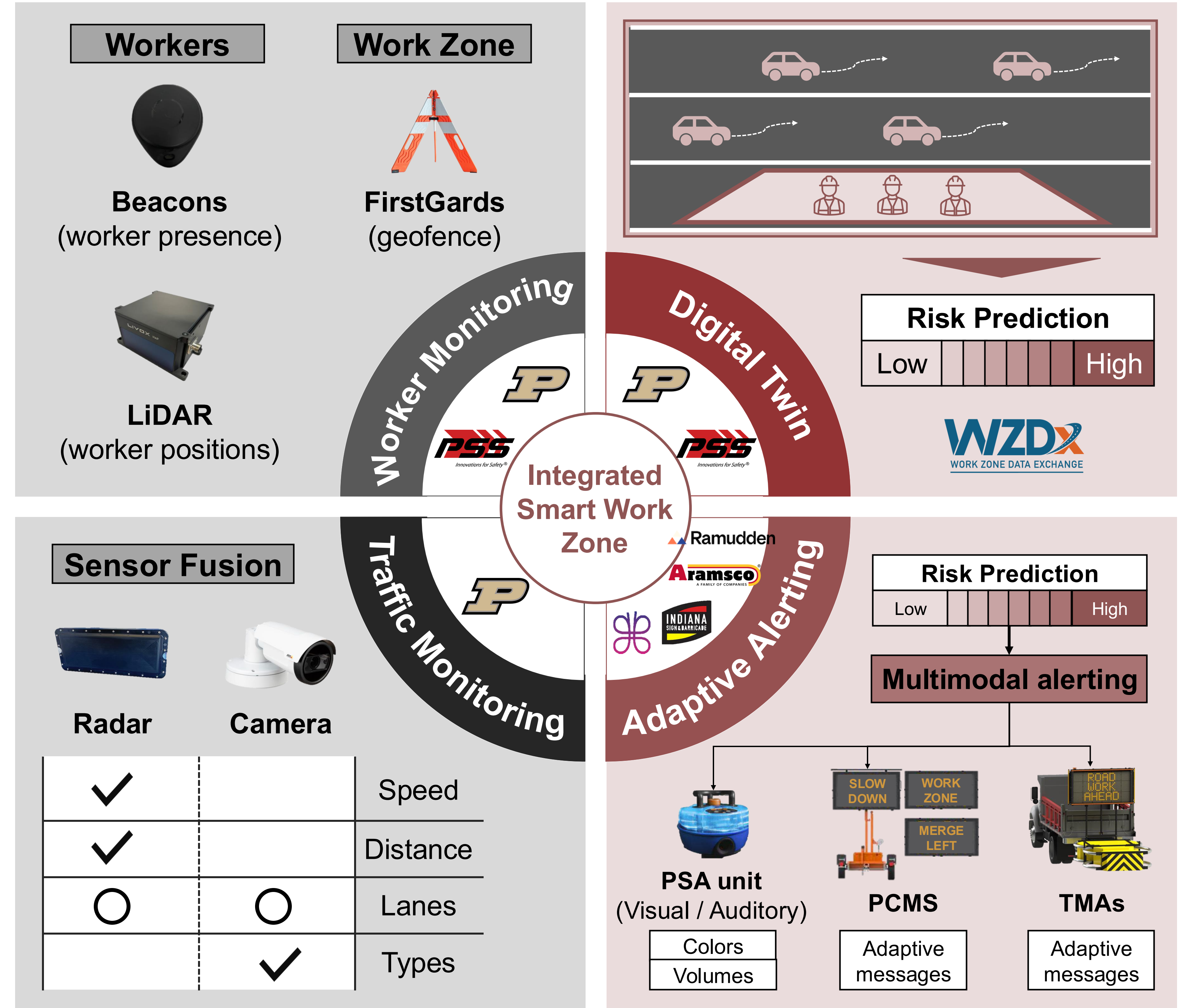
1. **No communication among tech.**

2. **Current and future traffic conditions**

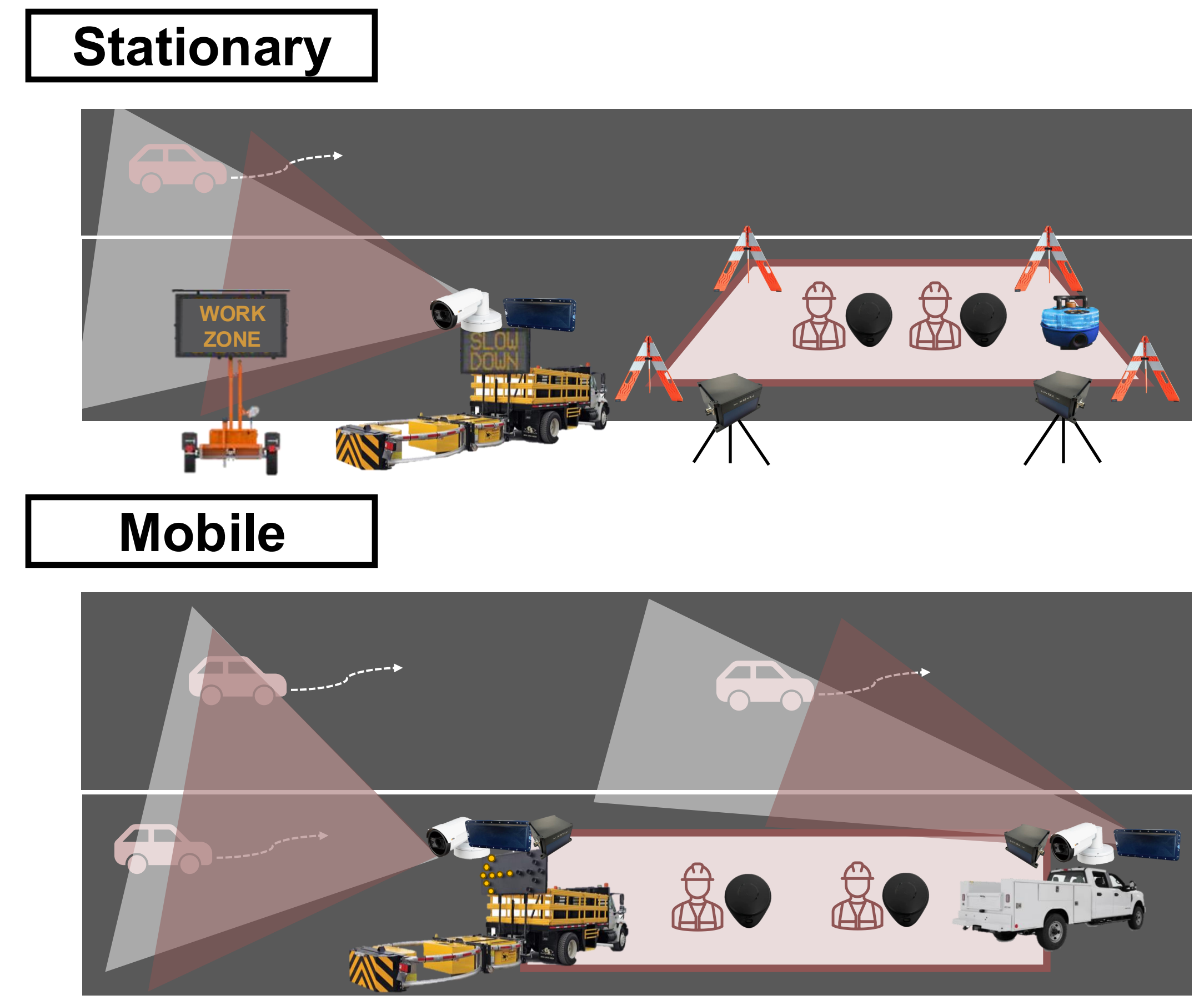
3. **Static and mobile work zones**

- Existing mitigation technologies did not address the **multi-dimensional** nature of the intrusions.
- An **integrated solution** is needed to thoroughly minimize work zone intrusions.

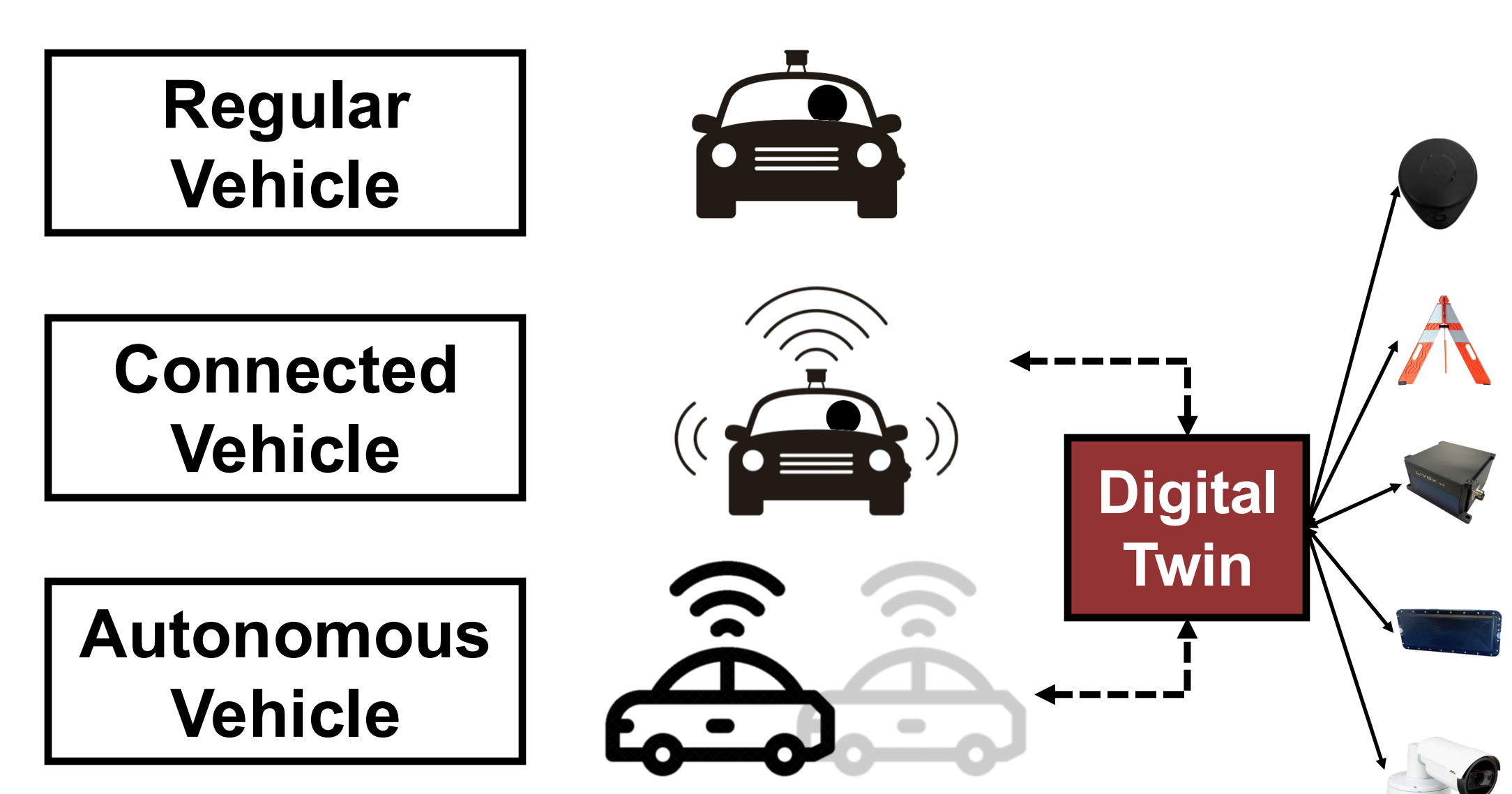
3. INTEGRATED SMART WORK ZONE



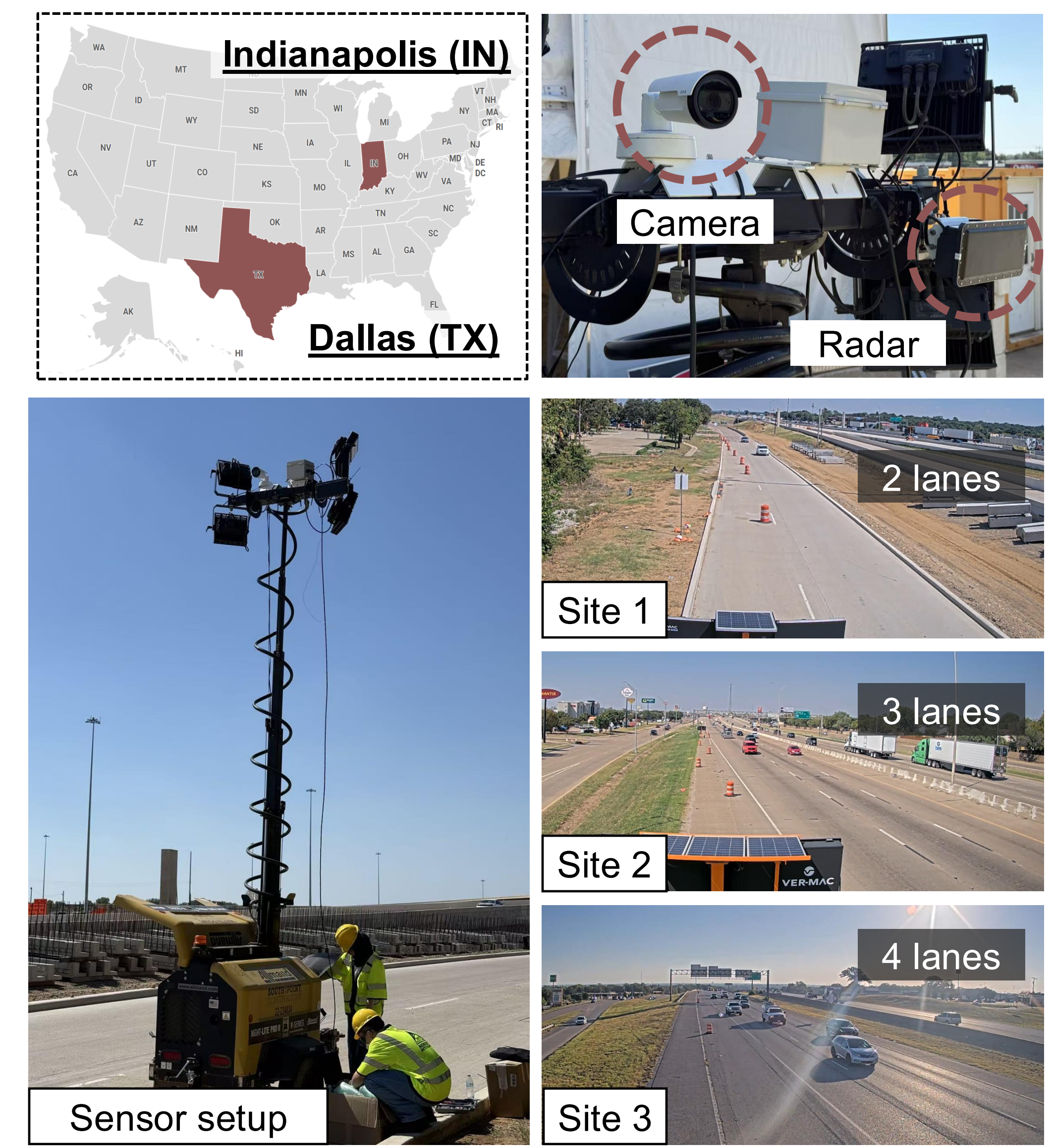
Work Zone Types



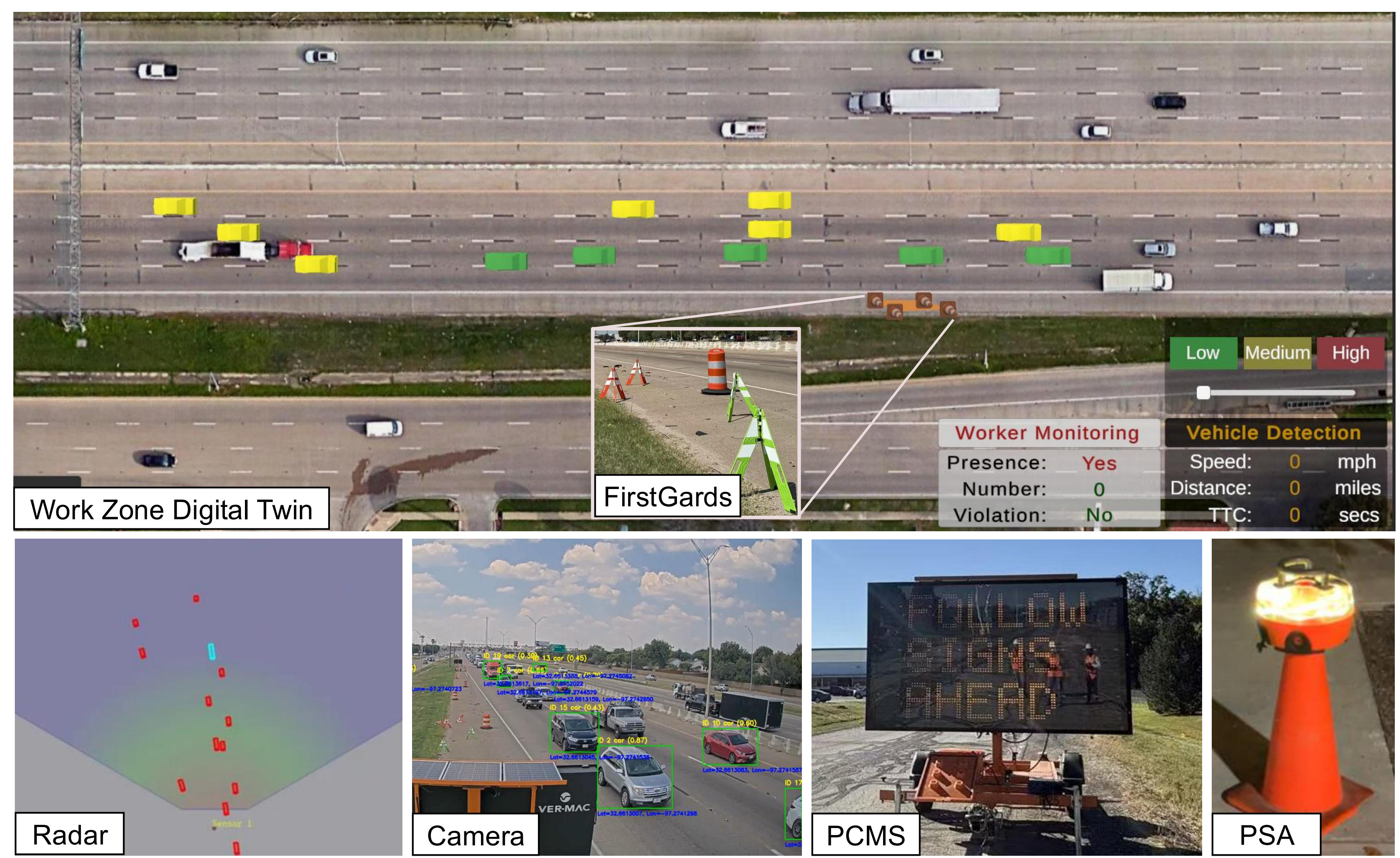
Vehicle Types



4. FIELD TESTS

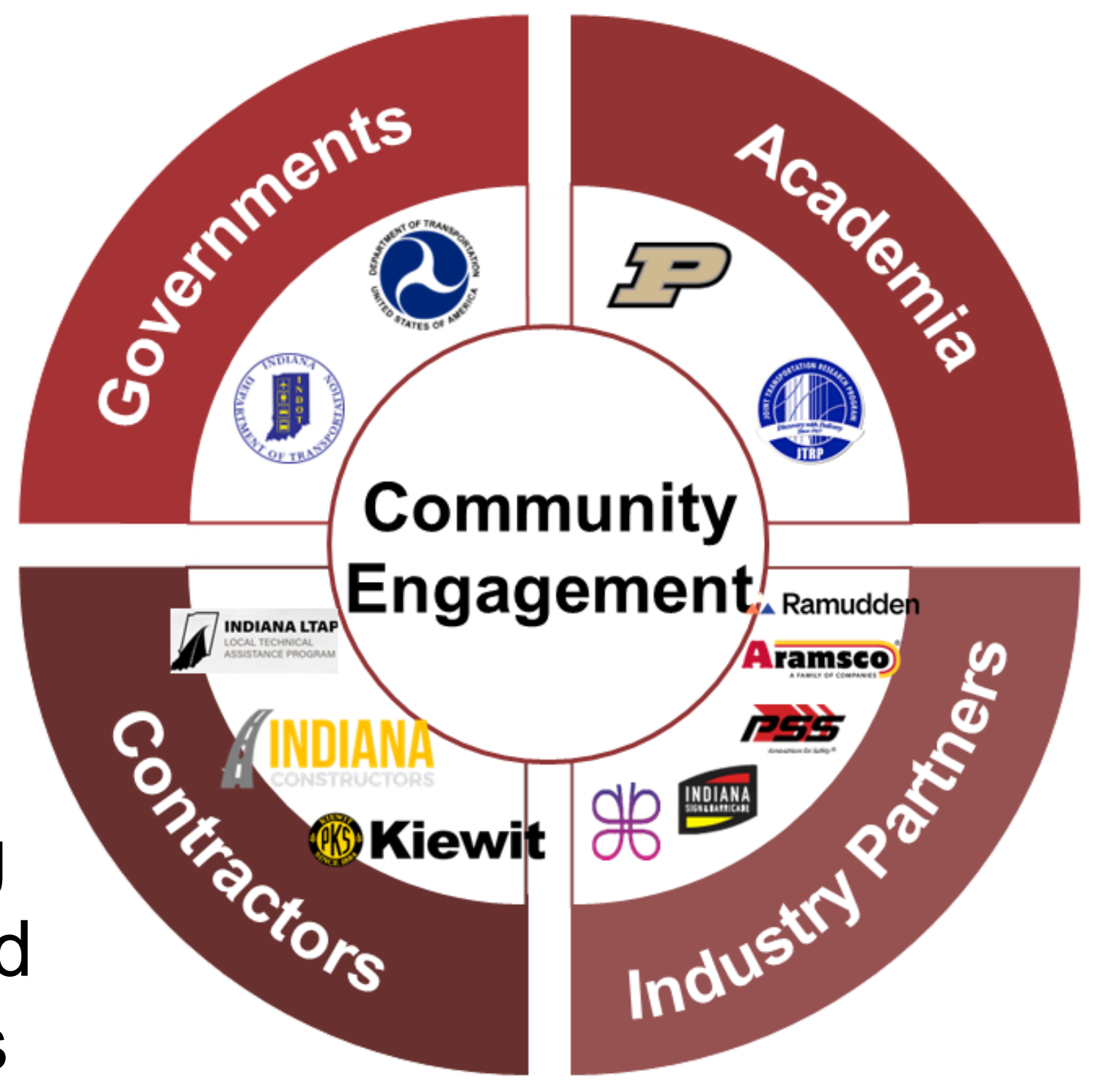


5. WORK ZONE DIGITAL TWIN DEMONSTRATION



6. CONCLUSIONS

- The **field test** showed that the digital twin can accurately **generate geofence**, **monitor workers**, **track traffic**, and **offer feedback**.
- Future work will include field tests across varying work zone conditions and connected / autonomous vehicles.



7. ACKNOWLEDGEMENT

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